

GURKEERAT JAWANDA

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EDUCATION

University of Canberra

Bachelor of Software Engineering (Robotics and AI Major) | GPA: 6.1/7
Dean's Excellence Award

Canberra, ACT
December 2025

EXPERIENCE

University of Canberra

Research Assistant, Power Grid Blackout Prediction

Canberra, ACT
February 2025 - July 2025

- Collaborated with a research team to build a power grid blackout prediction system using Logistic Regression, Random Forest, and K-Nearest Neighbors models.
- Trained and evaluated multiple classification models on historical grid outage data to identify patterns linked to blackouts.
- Validated model predictions across 90 test cases to assess the accuracy and reliability of the blackout prediction system.

PROJECTS

AI Helpdesk Automation System | FastAPI, Pydantic, SQLAlchemy

- Built an automated helpdesk system that streamlines support ticket intake, categorization, and routing using FastAPI.
- Designed structured data models and request validation schemas with Pydantic to ensure reliable API interactions.
- Implemented persistent ticket and user data storage with SQLAlchemy, connecting the application to a relational database.

ML Financial Market Analysis | SVM, XGBoost, PCA

- Built a financial market analysis pipeline using SVM and XGBoost models to classify and predict market trends from historical data.
- Applied Principal Component Analysis to reduce feature dimensionality and improve model training efficiency.
- Compared performance across multiple algorithms to identify the most effective approach for market trend prediction.

Android Real-Time Image Analysis App | Java, Kotlin, Firebase, Google ML Kit

- Developed an Android application that performs real-time image analysis using Google ML Kit.
- Built the app with Java and Kotlin, integrating Firebase for backend data storage and user management.
- Implemented on-device image processing to deliver real-time analysis results without relying on external APIs.

TECHNICAL SKILLS

- Programming Languages:** Python, Java, Kotlin, JavaScript/TypeScript, SQL, C, HTML/CSS
- Frameworks and Libraries:** FastAPI, Flask, Pydantic, SQLAlchemy, React, scikit-learn, PyTorch
- Machine Learning:** Logistic Regression, Random Forest, K-Nearest Neighbors, SVM, XGBoost, PCA
- Databases and Backend:** PostgreSQL, SQLite, Firebase
- Tools and Platforms:** Git, Docker, Linux, Android Studio, Google ML Kit